

June 17, 2026

Dear PhD Admissions Committee,

I am writing to express my strong interest in the PhD position at the University of Southampton, focusing on developing algorithms for molecular simulations on quantum computers. As an AI Data Science Engineer with a background in computational chemistry and a passion for machine learning, I believe this opportunity aligns perfectly with my research goals and interests.

My research experience has equipped me with a unique blend of skills that would be highly valuable in this project. For instance, my work on *MedicalMambaTS*, a linear – time state space model for sub – millisecond bedside inference, has taught me the importance of developing efficient algorithms that can process large amounts of data.

Furthermore, my experience with *PI_DeepONet*, a 0D Windkessel Physics – Informed Operator Network using PyTorch Autograd, has deepened my understanding of the intersection of machine learning and quantum computing.

I am particularly excited about the opportunity to explore the intersection of machine learning and quantum computing. The potential for fundamental changes in our capabilities to model materials, as well as the challenges presented by the need to adapt algorithm design to new architecture operating around superposition and entanglement, resonates deeply with me.

I am confident that my unique blend of skills and experience make me an ideal candidate for this PhD position. I look forward to discussing my application further.

Sincerely,